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Independent Investigative Analyst | Blockchain Forensics Transition | July 2026

Portfolio Disclosure

These work samples were produced during an intensive self-directed study of blockchain forensics methodology, beginning from near-zero cryptocurrency knowledge and progressing through live chain tracing, Python tool development, and formal case documentation. They represent original analytical work against real publicly available blockchain data, and are presented as evidence of investigative capability and learning velocity rather than credentialed professional output.

What This Portfolio Demonstrates

The case studies, methodology documents, and analytical tools contained in this portfolio were developed independently over a period of intensive self-directed study. Each element represents a genuine learning progression — from first principles through applied investigation against live blockchain data.

This portfolio is not presented as the output of a trained, credentialed forensic analyst. It is presented as evidence of the following:

Investigative Instinct — The ability to identify suspicious patterns in raw blockchain data without being told what to look for — recognizing structuring, fan-out architecture, dormancy periods, and round-number engineering through analytical reasoning rather than checklist application.

Technical Build Capability — Development of a seven-indicator Python red-flag scoring framework from scratch, calibrated through testing against real live wallet data including the Colonial Pipeline/DarkSide case, a sextortion network, an advance-fee scam, and a 94-day multi-layer laundering chain.

Methodology Development — Original analytical frameworks including an Alias Detection Framework, Retrograde Network Analysis approach, and Portnoff-derived timestamp correlation methodology — all developed independently and documented as formal white papers.

Learning Velocity — Progression from near-zero cryptocurrency knowledge to producing formal end-to-end blockchain trace reports, Python forensics tools, and methodology documentation within a compressed self-directed learning timeline.

AI-Augmented Analysis — Use of AI as an active investigative reasoning partner — pressure-testing hypotheses, synthesizing large volumes of data, and co-developing Python tools — demonstrating the hybrid human-AI analytical workflow that defines modern intelligence work.

Portfolio Contents

Case Studies

Case Study	Summary
Case Study 1 — Colonial Pipeline / DarkSide Ransomware	Full forward and backward trace of 75 BTC ransom payment through dormancy hops, CoinJoin mixing, and Kraken cash-out. Cross-referenced against published Chainalysis findings.
Case Study 2 — Sextortion Blackmail Network	Multi-layer fan-out structure traced from initial extortion payment through escalating distribution network terminating at high-volume custodial wallet.
Case Study 3 — Advance-Fee Bitcoin Scam	Statistical analysis confirming 93% deposit clustering around historically-verified \$25 target amount. All 86 victim payments swept in single coordinated transaction. CV analysis performed in Python.
Case Study 4 — Two-Track Parallel Laundering Chain	Initial documentation of the February 25, 2020 collection wallet and structured aggregator — later expanded into Case Study 5.
Case Study 5 — 94-Day Multi-Layer Laundering Operation	Complete end-to-end trace of a \$390,284+ operation spanning February 15 to May 20, 2020. Includes automated HD wallet generation, time-staggered consolidation, three dormancy periods, round-number output engineering, and dual cash-out at two exchange clusters.

Methodology White Papers

Alias Detection Framework — A systematic approach to identifying manufactured identities anchored by the census falsifiability test — surnames that do not appear in historical census data at statistically expected frequencies signal constructed identities.

Retrograde Network Analysis — A methodology for tracing backwards from known outcomes — arrests, convictions, registry entries — to map upstream actors, facilitators, and repeat-pattern offenders across jurisdictions.

Technical Tools

Bitcoin Wallet Red-Flag Scorer v3 — Python script implementing seven weighted indicators including drain-to-zero detection, CV-based deposit clustering, value-weighted sweep dominance, dormancy gap measurement, and mixing-service contact detection. Calibrated through testing against four real-world cases.

Phone Log Analysis Template — Python pipeline for carrier call and text log analysis including burst detection, odd-hour contact flagging, onset-silence analysis, and gap measurement. Reusable across multiple carrier log formats.

Certifications

PURSUIT Certification — Guardian Group Project 1591 — Certificate ID: 7b3cae7zid. Curriculum required identifying trafficking victims and perpetrators from a single escort advertisement using digital artifacts, linguistic patterns, image metadata, and platform-specific indicators.

Tools and Resources

WalletExplorer.com — Wallet clustering, transaction history, named service identification

mempool.space — Raw transaction data, UTXO analysis, block confirmation verification, visual transaction mapping

Chainabuse — Known malicious address cross-reference

CoinGecko / CoinMarketCap — Historical BTC price data for USD conversion at transaction dates

Python 3.12 — Custom red-flag scoring framework, statistical analysis, phone log pipeline

AI-assisted development — Hypothesis testing, Python co-development, analytical synthesis

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All blockchain data sourced from public ledgers. All analytical work is original and independent.